```
========================================
            RECIPE QUEST
========================================

========================================
         LOADING RECIPE DATA
========================================
Reading: recipes_01_to_50.txt
Reading: recipes_101_to_150.txt
Reading: recipes_151_to_200_indian.txt
Reading: recipes_201_to_250_indian.txt
Reading: recipes_251_to_300_foreign.txt
Reading: recipes_301_to_350_foreign.txt
Reading: recipes_51_to_100.txt

========================================
Total recipes loaded: 350
========================================

========================================
             MAIN MENU
========================================
1. Search Recipe
2. Max Flow
3. Knapsack FPTAS (CO5)
4. Exit
========================================
Enter your choice: paneer

Invalid choice. Please try again.

========================================
             MAIN MENU
========================================
1. Search Recipe
2. Max Flow
3. Knapsack FPTAS (CO5)
4. Exit
========================================
Enter your choice: 1

========================================
          RECIPE SEARCH
========================================
Enter recipe name: paneer

========================================
       SELECT SEARCH ALGORITHM
========================================
1. KMP
2. Rabin-Karp
3. Z-Function
4. Edit Distance
5. Compare All Algorithms
6. Randomized QuickSort (CO6)
7. Back to Main Menu
========================================
Enter algorithm choice: 1

========================================
             KMP SEARCH
========================================
Search Query: paneer
Searching recipes...

Match found at Recipe ID: 6
```

```
========================================
       SELECT SEARCH ALGORITHM
========================================
1. KMP
2. Rabin-Karp
3. Z-Function
4. Edit Distance
5. Compare All Algorithms
6. Randomized QuickSort (CO6)
7. Back to Main Menu
========================================
Enter algorithm choice: 1

========================================
            KMP SEARCH
========================================
Search Query: paneer
Searching recipes...

Match found at Recipe ID: 6

========================================
            RECIPE FOUND
========================================
Recipe ID : 6
Recipe    : Paneer Butter Masala
Cuisine   : North India
Category  : Main Course

PREPARATION STEPS:
----------------------------------------
1. 1. Cut paneer into cubes.
2. 2. Lightly fry paneer and keep it aside.
3. 3. Cook tomatoes with cashews, ginger, and garlic.
4. 4. Blend the cooked mixture smoothly.
5. 5. Heat butter and add the tomato paste.
6. 6. Add chilli powder, garam masala, salt, and water.
7. 7. Simmer until the gravy thickens.
8. 8. Add cream and mix gently.
9. 9. Add paneer and simmer briefly.
10. 10. Garnish with coriander.
========================================
----------------------------------------

Match found at Recipe ID: 7

========================================
            RECIPE FOUND
========================================
Recipe ID : 7
Recipe    : Palak Paneer
Cuisine   : North India
Category  : Main Course

PREPARATION STEPS:
----------------------------------------
1. 1. Wash spinach thoroughly.
2. 2. Blanch spinach briefly in hot water.
3. 3. Transfer spinach to cold water.
4. 4. Blend spinach with ginger, garlic, and green chilli.
5. 5. Cut paneer into cubes and lightly fry if desired.
6. 6. Cook cumin and chopped onion in oil.
7. 7. Add spinach puree and cook gently.
8. 8. Add salt, garam masala, and cream.
9. 9. Add paneer and simmer briefly.
10. 10. Serve with roti or rice.
```

```
=====================================
            MAIN MENU
=====================================
1. Search Recipe
2. Max Flow
3. Knapsack FPTAS (CO5)
4. Exit
=====================================
Enter your choice: 2

=====================================
       RECIPE SUPPLY NETWORK
=====================================

SOURCE -> VEGETABLES : 10
SOURCE -> MEAT       : 15
SOURCE -> SPICES     : 10
SOURCE -> RICE       : 8

VEGETABLES -> RECIPE : 7
MEAT -> RECIPE       : 10
SPICES -> RECIPE     : 6
RICE -> RECIPE       : 5

RECIPE -> SINK       : 20

=====================================
      EDMONDS-KARP MAX FLOW
=====================================
Augmenting Path 1 : SOURCE -> VEGETABLES -> RECIPE -> SINK
Path Flow       : 7
Current Max Flow : 7
-------------------------------------
Augmenting Path 2 : SOURCE -> MEAT -> RECIPE -> SINK
Path Flow       : 10
Current Max Flow : 17
-------------------------------------
Augmenting Path 3 : SOURCE -> SPICES -> RECIPE -> SINK
Path Flow       : 3
Current Max Flow : 20
-------------------------------------
Maximum Flow     : 20
=====================================

=====================================
         MAX FLOW SUMMARY
=====================================
Algorithm        : Edmonds-Karp
Maximum Flow     : 20
Execution Time   : 10169200 ns
Execution Time   : 10.1692 ms
=====================================

=====================================
            MAIN MENU
=====================================
1. Search Recipe
2. Max Flow
3. Knapsack FPTAS (CO5)
4. Exit
=====================================
Enter your choice:
```

```
  ========================================
         CO5 - KNAPSACK FPTAS
  ========================================
  Enter recipe search query: paneer

  Matching Recipes:
  ----------------------------------------
  1. Paneer Butter Masala (ID: 6)
  2. Palak Paneer (ID: 7)
  3. Shahi Paneer (ID: 8)
  4. Paneer Tikka (ID: 38)
  5. Paneer Biryani (ID: 158)
  6. Kadai Paneer (ID: 159)
  7. Matar Paneer (ID: 160)
  8. Paneer Bhurji (ID: 161)
  ----------------------------------------
  Enter the number of recipes to consider (1-8): 4

  Enter the recipe numbers you want to optimize.
  Recipe 1 number: 3
  Recipe 2 number: 4
  Recipe 3 number: 6
  Recipe 4 number: 7
  Enter maximum budget/capacity: 1000

  Some selected recipes do not have COST/VALUE in the TXT file.
  To use automatic Knapsack data, add these fields to each recipe:
  COST: <non-negative integer>
  VALUE: <non-negative integer>
```

```
========================================
       CO5 - KNAPSACK FPTAS
========================================
Enter recipe search query: paneer

Matching Recipes:
----------------------------------------
1. Paneer Butter Masala (ID: 6)
2. Palak Paneer (ID: 7)
3. Shahi Paneer (ID: 8)
4. Paneer Tikka (ID: 38)
5. Paneer Biryani (ID: 158)
6. Kadai Paneer (ID: 159)
7. Matar Paneer (ID: 160)
8. Paneer Bhurji (ID: 161)
----------------------------------------
Enter the number of recipes to consider (1-8): 4

Enter the recipe numbers you want to optimize.
Recipe 1 number: 3
Recipe 2 number: 4
Recipe 3 number: 6
Recipe 4 number: 7
Enter maximum budget/capacity: 1000
```

```
========================================
          FPTAS RESULT
========================================
Maximum Budget : 1000

Selected Recipes:
----------------------------------------
Kadai Paneer (ID: 159)
   Cost  : 500
   Value : 200
----------------------------------------
Total Cost       : 500
Total Value      : 200
Budget Remaining : 500
Algorithm        : Knapsack FPTAS
Approximation ε  : 0.25
========================================

========================================
              MAIN MENU
========================================
1. Search Recipe
2. Max Flow
3. Knapsack FPTAS (CO5)
4. Exit
========================================
Enter your choice:
```

```
========================================
        SELECT SEARCH ALGORITHM
========================================
1. KMP
2. Rabin-Karp
3. Z-Function
4. Edit Distance
5. Compare All Algorithms
6. Randomized QuickSort (CO6)
7. Back to Main Menu
========================================
Enter algorithm choice: 4

========================================
            EDIT DISTANCE
========================================
Search Query: paneer
Finding closest recipe name...

========================================
          EDIT DISTANCE RESULT
========================================
Records Checked : 350
Closest Recipe  : Kheer
Recipe ID       : 43
Edit Distance   : 3
Execution Time  : 2892700 ns
Execution Time  : 2.8927 ms
========================================
```

```
Enter algorithm choice: 5

========================================
       STRING ALGORITHM COMPARISON
========================================
Search Query: paneer
Total Recipes: 350

----------------------------------------
KMP
----------------------------------------
Records Checked : 350
Matches Found   : 8
Execution Time  : 314200 ns
Execution Time  : 0.3142 ms

----------------------------------------
RABIN-KARP
----------------------------------------
Records Checked : 350
Matches Found   : 8
Execution Time  : 779500 ns
Execution Time  : 0.7795 ms

----------------------------------------
Z-FUNCTION
----------------------------------------
Records Checked : 350
Matches Found   : 8
Execution Time  : 5547900 ns
Execution Time  : 5.5479 ms

----------------------------------------
EDIT DISTANCE
----------------------------------------
Records Checked : 350
Closest Recipe  : Kheer
Recipe ID       : 43
Edit Distance   : 3
Execution Time  : 595000 ns
Execution Time  : 0.595 ms

========================================
          MATCHING RECIPES
========================================

Match found at Recipe ID: 6

========================================
            RECIPE FOUND
========================================
Recipe ID : 6
Recipe    : Paneer Butter Masala
Cuisine   : North India
Category  : Main Course
```